

An Introduction to L^AT_EX

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Cover

Outline

- 1 Introduction
 - What is L^AT_EX and why?
- 2 Using L^AT_EX
 - The basics
 - The basics of formatting
- 3 Document structure
- 4 Lists and tables
- 5 Mathematics
- 6 Figures and chemistry
- 7 References

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Why L^AT_EX?

If you merely want to produce a passably good document — something acceptable and basically readable but not really beautiful — a simpler system will usually suffice.

With T_EX the goal is to produce the finest quality.

—Donald E. Knuth

Why L^AT_EX?

L^AT_EX is a typesetting system (based on T_EX) designed for structured, technical writing.

For a bachelor's thesis that usually means:

- consistent layout across a long document
- strong support for formulas, figures, tables, and references
- easy automatic numbering and cross-referencing
- source files that stay manageable as the document grows

Why not just use Word?

- Word is often faster for short, simple documents
- L^AT_EX becomes attractive when you need:
 - many sections, figures, tables, and citations
 - stable numbering and references
 - precise control over formulas and notation
 - collaboration around plain-text source files
- It is not WYSIWYG
 - you write source code
 - then compile it to PDF

Workshop workflow

- We will assume you use Overleaf
 - no installation needed
 - editor, compiler, and PDF preview in one place
- Typical workflow:
 - start from a template or minimal file
 - edit the source on the left
 - recompile and inspect the PDF on the right
 - fix warnings and errors as you go
- For this workshop: focus on readable source, not perfect memorisation

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A minimal thesis-style document

- A document has two main parts:
 - the preamble: setup
 - the document body: content
- Typical starting points:
 - `\documentclass{article}`
 - `\usepackage{...}`
 - `\begin{document} ... \end{document}`
- For a thesis, the document class is often given by your programme, faculty, or publisher

Commands, environments, and sections

- Commands start with a backslash: `\title{...}`,
`\textbf{...}`
- Many commands take arguments in braces
- Environments start and end in pairs: `\begin{figure} ...`
`\end{figure}`
- Structure matters more than appearance:
 - `\section{Introduction}`
 - `\subsection{Methods}`
 - `\tableofcontents`
- Numbering and contents entries are generated automatically

Text flow and input

- L^AT_EX automatically handles:
 - spacing
 - paragraph layout
 - hyphenation
 - numbering of sections, figures, tables, and equations
- White space is mostly ignored
- An empty line starts a new paragraph
- % starts a comment

Special characters and text formatting

- These characters have a special meaning: # \$ % ^ & _ { } ~ \
- Use formatting commands instead of manual styling:
 - `\textbf{...}` for bold
 - `\emph{...}` for emphasis
 - `\texttt{...}` for code-like text
- A few common details:
 - quotes are typed as “like this”
 - hyphen, en-dash, em-dash: -, --, ---
 - `\ldots` gives an ellipsis: ...

Some useful formatting commands

<code>\textbf{...}</code>	bold text
<code>\textit{...}</code>	italic text
<code>\emph{...}</code>	semantic emphasis
<code>\underline{...}</code>	underlined text
<code>\texttt{...}</code>	monospaced text
<code>\small</code>	smaller text size
<code>\Large</code>	larger text size

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Document structure for a thesis

- Front matter often contains:
 - title page
 - abstract
 - table of contents
- Main matter is built from sections and subsections
- Back matter may contain appendices and the bibliography
- Good structure makes later editing much easier than manual formatting does

Cross-references

- Let L^AT_EX number things for you
- Add a label where something is defined:
`\label{sec:method}`
- Refer to it later with:
 - `\ref{sec:method}`
 - `\pageref{sec:method}`
- This works for sections, equations, figures, and tables
- Result: fewer broken references after editing

Bibliography

- Thesis writing almost always means reference management
- Keep references in a separate .bib file
- Cite in the text with a command such as `\cite{...}`
- Print the reference list with your chosen bibliography setup
- The exact citation style is usually decided by your programme, supervisor, or journal

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Lists

- Three common list environments:
 - `itemize` for bullet points
 - `enumerate` for numbered steps
 - `description` for term-definition lists
- Each new list item starts with `\item`
- Lists can be nested
- Useful for protocols, assumptions, and summarising results

Tables

- Use `tabular` to align values in columns
- Put important tables in a `table` environment if they need a caption and label
- Keep table structure simple and readable
- For scientific tables, consistency of units and significant figures matters more than decorative lines

Sample	Yield (%)
A	81
B	77
C	84

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Math mode

- L^AT_EX has a special mode for mathematical notation
- Common forms:
 - inline math with $\dots$
 - display math with
$$\dots$$
 - numbered equations with `equation`
- Superscripts and subscripts are built in
- This matters even in chemistry theses for symbols, variables, and models

Examples

Inline example: $c^2 = a^2 + b^2$

Displayed and numbered:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}$$

Useful commands include `\frac`, `\sqrt`, superscripts with `^`, and subscripts with `_`

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Figures and floats

- Use `\usepackage{graphicx}` to include images
- Put important figures in a `figure` environment
- Add a `\caption` and `\label` so you can refer to them later
- Let L^AT_EX help with placement instead of forcing every image by hand



Figure: Example figure placeholder

Chemistry: formulas and reactions

- A useful package for chemistry is `mhchem`
- It provides `\ce{...}` for formulas, charges, and reactions
- Examples:
 - H_2O
 - Fe^{3+}
 - $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$
 - $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$
- This is usually clearer and more reliable than hand-formatting chemistry notation

Chemistry: units and measurements

- A useful package for quantities is `siunitx`
- It keeps values and units consistent throughout a thesis
- Examples:
 - `\SI{25}{\milli\liter}` gives 25 mL
 - `\SI{298.15}{\kelvin}` gives 298.15 K
 - `\num{6.022e23}` gives 6.022×10^{23}
- This is especially useful in result tables and experimental sections

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Further reading

- Overleaf documentation:
<https://www.overleaf.com/learn>
- The Not So Short Introduction to L^AT_EX:
<https://ctan.org/pkg/lshort>
- CTAN package archive: <https://ctan.org>
- Wikibooks pages on tables, mathematics, and bibliography management

Summary

- For your longer work (e.g. theses), focus on:
 - structure before styling
 - labels and references instead of manual numbering
 - bibliography management from the start
 - packages only when they solve a real problem
- For chemistry, remember the package names `mhchem` and `siunitx`

Getting started...

- Open the workshop exercise file: t.ly/iyYsG
- Experiment! Feel free to extend or change the exercises as you like.
- Ask for help if you get stuck. Feel free to use Google or your favorite LLM.